

Multidimensional subscript operator

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Abstract

We propose that user-defined types can define a subscript operator with multiple arguments to better support multi-dimensional containers and views.

Tony tables

Before	After
<pre>template<class ElementType, class Extents> class mdpan { template<class... IndexType> constexpr reference operator()(IndexType...); }; int main() { int buffer[2*3*4] = { }; auto s = mdspan<int, extents<2, 3, 4>>(buffer); s(1, 1, 1) = 42; }</pre>	<pre>template<class ElementType, class Extents> class mdpan { template<class... IndexType> constexpr reference operator[](IndexType...); }; int main() { int buffer[2*3*4] = { }; auto s = mdspan<int, extents<2, 3, 4>>(buffer); s[1, 1, 1] = 42; }</pre>

Revisions

R2

- Add explanation about not adapting this proposal to C arrays
- Remove the restriction to require at least one parameter

- Add a paragraph about valarray

Motivation

Types that represent multidimensional views (`mdspan`), containers (`mdarray`), grid, matrixes, images, geometric spaces, etc, need to index multiple dimensions.

In the absence of a more suitable solution, these classes overload the call operator. While this is functionally equivalent to the proposed multidimensional subscript operator, it does not carry the same semantic, making the code harder to read and reason about. It also encourages non-semantical operator overloading.

Proposal

We propose that the `operator[]` should be able to accept 0 or more arguments, including variadic arguments. Both its use and definition would match that of `operator()`, with the exception that at least one argument would be required.

We make the expressions deprecated in 20 ill-formed while allowing multi-dimensional subscripts expressions in **new** standard types and user types. We do not propose modifications to C arrays as to leave a cycle before giving new meaning to syntax that was still valid in C++20.

What about comma expressions?

In C++20 we deprecated the use of comma expressions in subscript expressions [P1161R3][?]. This proposal would make these ill-formed and give a new meaning to commas in subscript expressions. While the timeline is aggressive, we think it is important that this feature be available for the benefit of `mdspan` and `mdarray`. At the time of writing [P1161R3], [?] has been implemented by at least GCC, clang, and MSVC. [P1161R3][?] further denotes that the cases where comma expressions appear in subscript are vanishingly rare.

However, an implementation could keep supporting the current behavior as an extension, for example, they could fall-back to a comma expression if no overload is found for an expression list, or always assume a comma expression in the presence of a C-array.

Because we should not make C++ more confusing, we think the standard should not continue to support the old meaning of a comma in subscript expressions.

What about `[foo][bar]`?

As mentioned in [P1161R3][?], an `operator[]` can return an object which has itself an `operator[]`. Therefore chaining multiple `[]` to index a single object isn't a viable proposal.

Should we adopt the same syntax for C arrays?

Code that is deprecated in 20, should be ill-formed in 23 rather than a potentially silent change. As such we do not propose the proposed syntax to apply to C arrays. The usefulness of this should be discussed in the C++26 time frame. However C arrays are not widely used by C++, spending time on them might therefore not be useful.

Should we add a multidimensional operator to valarray?

Again, we shouldn't change the meaning of existing code in C++23. We should only add multidimensional operators to new in C++23 types such as `mdspan`. If there are users of `valarray` interested in this feature, this can be done in C++26

Wording



Expressions

[expr]



Postfix expressions

[expr.post]

Postfix expressions group left-to-right.

postfix-expression: *primary-expression* ~~*postfix-expression* [*expr-or-braced-init-list*]~~
postfix-expression [*expression-list*] *postfix-expression* [*braced-init-list*] *postfix-expression* (*opt**expression-list*) *simple-type-specifier* (*opt**expression-list*) *typename-specifier* (*opt**expression-list*) *simple-type-specifier* *braced-init-list*



Subscripting

[expr.sub]

A postfix expression followed an expression in square brackets is a postfix expression. One of the expressions shall be a glvalue of type "array of T" or a prvalue of type "pointer to T" and the other shall be a prvalue of unscoped enumeration or integral type. The result is of type "T". The type "T" shall be a completely-defined object type.¹ The expression `E1[E2]` is identical (by definition) to `*((E1)+(E2))`, except that in the case of an array operand, the result is an lvalue if that operand is an lvalue and an xvalue otherwise. The expression `E1` is sequenced before the expression `E2`.

[Note: A comma expression appearing as the *expr-or-braced-init-list* of a subscripting expression is deprecated; see [depr.comma.subscript]. — end note]

[Note: Despite its asymmetric appearance, subscripting is a commutative operation except for sequencing. See [expr.unary] and [expr.add] for details of `*` and `+` and [dcl.array] for details of array types. — end note]

¹This is true even if the subscript operator is used in the following common idiom: `&x[0]`.

~~A *braced-init-list* shall not be used w~~ With the built-in subscript operator, a *braced-init-list* shall not be used and a *expression-list* shall be a single expression.

❖ **Overloaded operators** [over.oper]

❖ **Subscripting** [over.sub]

A subscripting operator function is a function named `operator[]` that is a non-static member function ~~with exactly one parameter~~. For an expression of the forms

postfix-expression [*expr-or-braced-init-list*]

postfix-expression [*expr-or-braced-init-list*]

postfix-expression [*expression-list*]

the operator function is selected by overload resolution ([over.match.oper]). If a member function is selected, the expression is interpreted as

the operator function is selected by overload resolution (xref). If a member function is selected, the expression is interpreted, respectively, as

postfix-expression . *operator* [] (*expr-or-braced-init-list*)

postfix-expression . *operator* [] (*expression-list*)

postfix-expression . *operator* [] (*braced-init-list*)

[Example:

```
struct X {
    Z operator[](std::initializer_list<int>);
    Z operator[](auto...);
};
X x;
x[{1,2,3}] = 7;           // OK: meaning x.operator[]({1,2,3})
x[1,2,3] = 7;             // OK: meaning x.operator[](1,2,3)
int a[10];
a[{1,2,3}] = 7;           // error: built-in subscript operator
a[1,2,3] = 7;             // error: built-in subscript operator
```

— end example]

❖ **Comma operator** [expr.comma]

In contexts where comma is given a special meaning, [Example: in lists of arguments to functions ([expr.call]), subscript expressions and lists of initializers ([decl.init]) — end example] the comma operator as described in this subclause can appear only in parentheses. [Example:

f(a, (t=3, t+2), c);

has three arguments, the second of which has the value 5. — end example]

[*Note: A comma expression appearing as the *expr-or-braced-init-list* of a subscripting expression [expr.sub] is deprecated; see depr.comma.subscript. — end note*]

◆ C++ and ISO C++ 2020

[diff.cpp20]

◆ [expr.sub]: declarations

[diff.cpp20.expr.sub]

Change: Change the meaning of comma in subscript expressions.

Rationale: Enable repurposing a deprecated syntax to support multidimensional indexing.

Effect on original feature: Valid C++ program that uses a comma expression within a subscript expression may fail to compile.

`arr[1, 2]` //was equivalent to `arr[(1, 2)]`, now equivalent to `arr.operator[](1, 2)` or ill-formed

◆ Comma operator in subscript expressions[depr.comma.subscript]

A comma expression appearing as the *expr-or-braced-init-list* of a subscripting expression is deprecated. [*Note: A parenthesized comma expression is not deprecated. — end note*]

[*Example:*

```
void f(int *a, int b, int c) {  
    a[b,c];           // deprecated  
    a[(b,c)];         // OK  
}
```

— end example]

Implementation

A prototype has been implemented in Clang.

[Compiler Explorer Demo](#).

Github: <https://github.com/cor3ntin/llvm-project/tree/subscript>

Acknowledgments

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References

[N4861] Richard Smith *Working Draft, Standard for Programming Language C++*
<https://wg21.link/N4861>